

Laboratorio 2

Anna Carelli (1029621) e Ilaria Consoli (944548)

Introduzione

Il laboratorio consiste nel confrontare le definizioni presenti nel database e quelle dei synset di WordNet (limitato ai sostantivi), con l'obiettivo di trovare un match tra ogni concetto definito e il corretto significato. Abbiamo pensato a due approcci per il criterio di comparazione:

- Confronto per similarità lessicale: calcolo del grado di sovrapposizione con Jaccard
- Confronto per similarità semantica: calcolo del cosine similarity tra gli embedding delle due definizioni.

Strumenti utilizzati

Le librerie utilizzate sono le seguenti:

- *nltk*
 - *nltk.corpus* per accedere a WordNet (wordnet as wn)
 - *nltk.corpus* e *nltk.stem* per il preprocessing (stopwords e WordNetLemmatizer)
- *pandas* e *numpy* per calcoli di supporto e manipolazione delle strutture dati
- *spacy* per il calcolo degli embedding
- *matplotlib.pyplot* per la realizzazione dei grafici di analisi

Implementazione

Approccio lessicale

Implementiamo un approccio lessicale per identificare il synset di WordNet che meglio rappresenta la definizione in input. Il codice itera attraverso tutti i synset classificati come sostantivi presenti nel database. Per ognuno, estrae la definizione testuale, ne pre-processa il testo. Utilizza poi la funzione jaccard per calcolare il grado di sovrapposizione tra l'input e la definizione del synset. La funzione mantiene traccia del punteggio massimo e del relativo synset, restituendoli infine come risultato dell'algoritmo.

Python

```
def synsets(string):  
    best_syn = None  
    best_overlap = 0  
  
    for syn in wn.all_synsets(pos='n'):  
        syn_text = syn.definition()  
        syn_words = nltk.word_tokenize(syn_text)
```

```

        filtered_syn = set([word for word in syn_words if word.isalpha() and
word not in stop_words])
        overlap = jaccard(string, " ".join(filtered_syn))

        if overlap > best_overlap:
            best_overlap = overlap
            best_syn = syn
    return best_syn, best_overlap

```

Approccio Semantico

Prima di eseguire il vero e proprio confronto, utilizziamo la seguente funzione per convertire e salvare gli embedding di ogni synset contenuto in WordNet.

Il codice esegue un'iterazione su tutti i synset, trasformando ognuno in un oggetto Doc di SpaCy dotato di corrispondente vettore numerico, a partire dalla combinazione della definizione e i nomi dei lemmi. I vettori e i relativi synset vengono infine memorizzati in liste parallele (*synset_vectors* e *valid_synsets*) per consentire calcoli di similarità semantica veloci durante la fase di ricerca.

```

Python
nlp = spacy.load("en_core_web_lg")

def setup_wordnet():
    synsets_list = list(wn.all_synsets(pos='n'))
    synset_vectors = []
    valid_synsets = []

    for s in synsets_list:
        content = s.definition() + " " + " ".join(s.lemma_names())
        vec = nlp.make_doc(content)

        if vec.has_vector and vec.vector_norm > 0:
            synset_vectors.append(vec)
            valid_synsets.append(s)

    return synset_vectors, valid_synsets

```

Dopodichè, per il confronto effettivo viene chiamata la funzione *synsets2*. Questa trasforma la stringa di input in un embedding valido (come visto precedentemente). Successivamente, la funzione itera attraverso la lista dei vettori pre-calcolati, calcolando la similarità tra l'input e ogni candidato tramite il metodo *similarity*. Tiene traccia del synset che presenta la maggiore vicinanza semantica nello spazio vettoriale, per poi restituirlo alla fine.

Python

```
def synsets2(string, synset_vectors, valid_synsets):
    emb_string = nlp.make_doc(string)

    if not emb_string.has_vector or emb_string.vector_norm == 0:
        return None, 0

    best_syn = None
    best_sim = 0

    for i in range(len(synset_vectors)):
        sim = emb_string.similarity(synset_vectors[i])
        if sim > best_sim:
            best_syn = valid_synsets[i]
            best_sim = sim

    return best_syn, best_sim
```

Difficoltà riscontrate

La complicazione maggiore deriva dalla mole di synset da confrontare con ogni l'input. Inizialmente per il calcolo degli embedding abbiamo impiegato l'intera pipeline di spaCy (*nlp*), ma l'esecuzione risultava troppo lunga. Fortunatamente, abbiamo risolto il problema limitandoci ad usare una funzione più veloce che calcolasse solo il vettore (*make_doc*) evitando gli altri passaggi superflui.

Risultati

Ecco i risultati ottenuti elencati in corrispondenza di ogni definizione fornita. Accanto a ogni synset è presente lo score di similarità ottenuto con Jaccard (JS) per il caso lessicale e con la Cosine Similarity (CS) per il caso semantico.

	Definizione fornita	Synset ottenuto (lex)	JS	Synset ottenuto (sem)	CS
1	A feeling of discomfort generated by embarrassing statements or behaviors. The term was coined on the Internet by the millennial community	mischief.n.01	0.12	visual_field.n.01	0.92
		reckless or malicious behavior that causes discomfort or annoyance in others		all of the points of the physical environment that can be perceived by a stable eye at a given moment	
2	Abstract mental content and result of mental thinking activity.	formation.n.07	0.29	cognition.n.01	0.89
		creation by mental activity		the psychological result of perception and learning and reasoning	
3	Interactive product of the	contraception.n.01	0.12	work.n.05	0.94

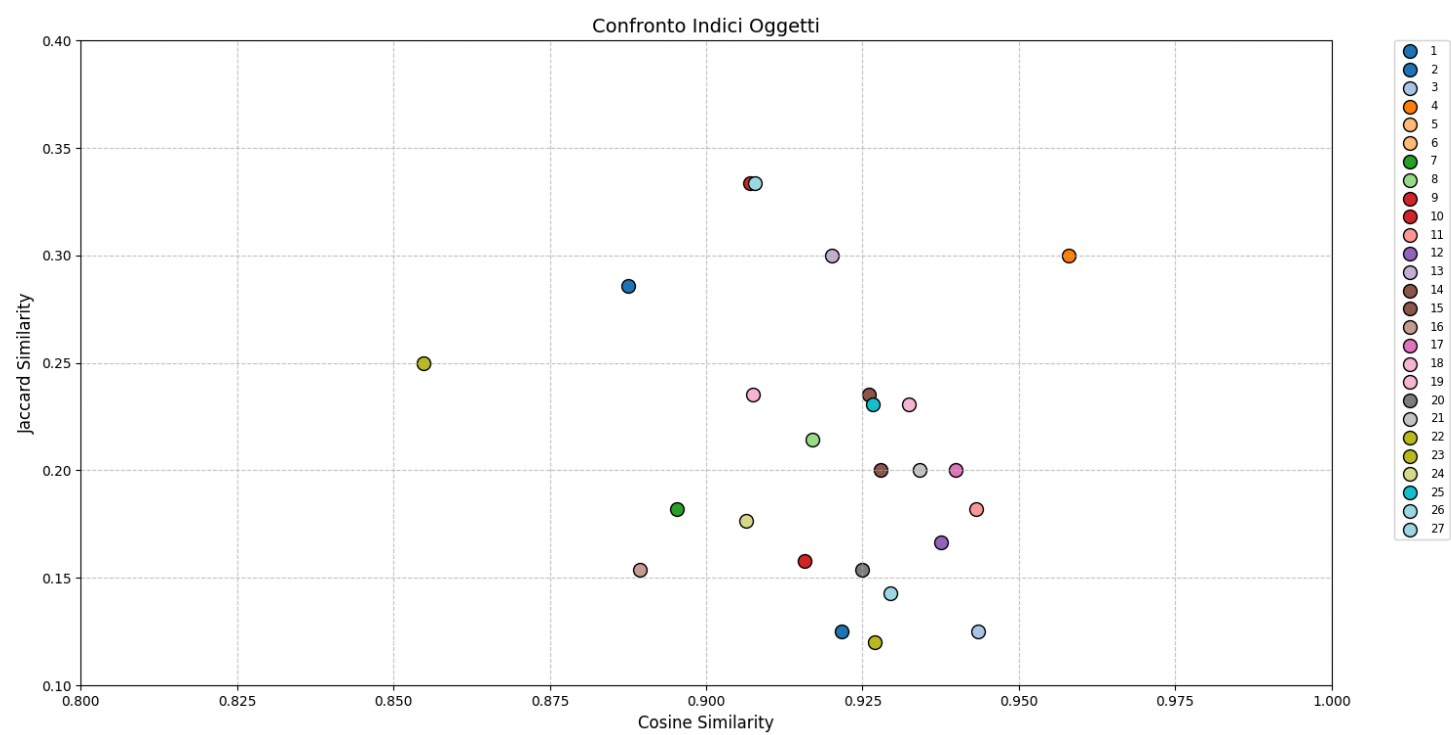
	Definizione fornita	Synset ottenuto (lex)	JS	Synset ottenuto (sem)	CS
	human intellect and creativity. Storydriven way of entertainment in which you take the control of the story with the use of an external device like a gamepad or a joystick.	birth control by the use of devices (diaphragm or intrauterine device or condom) or drugs or surgery		(physics) a manifestation of energy; the transfer of energy from one physical system to another expressed as the product of a force and the distance through which it moves a body in the direction of that force	
4	Feeling of discomfort during which one would like to be different from how one is	brotherhood.n.03	0.30	referred_pain.n.01	0.96
		the feeling that men should treat one another like brothers		pain that is felt at a place in the body different from the injured or diseased part where the pain would be expected	
5	Engineering system that has the purpose of capturing, purifying, transporting and distributing water	rediffusion.n.01	0.18	electrical_engineering.n.01	0.90
		a system for distributing radio or tv programs		the branch of engineering science that studies the uses of electricity and the equipment for power generation and distribution and the control of machines and communication	
6	Typical Japanese dish consisting mainly of rice and raw fish. Ability to add various toppings	sushi.n.01	0.21	rijsttaffel.n.01	0.92
		rice (with raw fish) wrapped in seaweed		dish originating in Indonesia; a wide variety of foods and sauces are served with rice	
7	Object used to improve vision through the use of graduated lenses	condenser.n.04	0.33	holography.n.01	0.91
		lens used to concentrate light on an object		the branch of optics that deals with the use of coherent light from a laser in order to make a hologram that can then be used to create a three-dimensional image	
8	A black substance produced from organic remains, flammable, and extractable from the subsoil. It sustains humanity on an energy level.	resorption.n.01	0.16	helium.n.01	0.92
		the organic process in which the substance of some differentiated structure that has been produced by the body undergoes lysis and assimilation		a very light colorless element that is one of the six inert gasses; the most difficult gas to liquefy; occurs in economically extractable amounts in certain natural gases (as those found in Texas and Kansas)	
9	This job is considered really important to the progression of the world	job_candidate.n.01	0.18	reality.n.03	0.94

	Definizione fornita	Synset ottenuto (lex)	JS	Synset ottenuto (sem)	CS
	and helps the make society progress	an applicant who is being considered for a job		the state of the world as it really is rather than as you might want it to be	
10	A series of smartphones designed and developed by Google at both the software and hardware levels.	cyberart.n.01	0.17	cyberart.n.01	0.94
		art that is produced with the help of computer hardware and software		art that is produced with the help of computer hardware and software	
11	A strong negative feeling or emotion towards another person that wronged you	emotional_person.n.01	0.30	anger.n.01	0.92
		a person subject to strong states of emotion		a strong emotion; a feeling that is oriented toward some real or supposed grievance	
12	Virtual or physical content that has the purpose of promoting goods, ideals or services	healthfulness.n.01	0.20	standard_of_living.n.01	0.93
		the quality of promoting good health		a level of material comfort in terms of goods and services available to someone or some group	
13	A piece of furniture featuring a comfortable, padded seat used for relax. It usually has two or more seats	kleenex.n.01	0.24	table.n.02	0.93
		a piece of soft absorbent paper (usually two or more thin layers) used as a disposable handkerchief		a piece of furniture having a smooth flat top that is usually supported by one or more vertical legs	
14	Pathological addiction, there are dedicated places generally prohibited to minors under 18 years of age.	hog.n.02	0.15	age_of_consent.n.01	0.89
		a sheep up to the age of one year; one yet to be sheared		the minimum age for marrying without parental consent or the minimum age for consensual sexual relations; intercourse at an earlier age can result in a charge of assault or statutory rape; the age differs in different states of the Union	
15	Small iron bar equipped with a hand grip and a flat or cross shaped tip used to tighten the screws	handlebar.n.01	0.20	machine_bolt.n.01	0.94
		the shaped bar used to steer a bicycle		a bolt with a square or hexagonal head on one end and a threaded shaft on the other end; tightened with a wrench; used to connect metal parts	
16	Tool used to unstuck tubes or pipes using a suction cup.	plunger.n.03	0.24	plunger.n.03	0.91
		hand tool consisting of a stick with a rubber suction cup at		hand tool consisting of a stick with a rubber suction cup at one	

	Definizione fornita	Synset ottenuto (lex)	JS	Synset ottenuto (sem)	CS
		one end; used to clean clogged drains		end; used to clean clogged drains	
17	Accessory for holding and gathering hair, often made of plastic with a spring in the center	sack.n.01	0.23	snood.n.01	0.93
		a bag made of paper or plastic for holding customer's purchases		an ornamental net in the shape of a bag that confines a woman's hair; pins or ties at the back of the head	
18	A key hormone that triggers a fight-or-flight response, preparing the body to react swiftly to the situation.	dressage.n.01	0.15	presence_of_mind.n.01	0.92
		maneuvers of a horse in response to body signals by the rider		self-control in a crisis; ability to say or do the right thing in an emergency	
19	Point of interaction with a more or less complex hardware system.	node.n.04	0.20	computer_system.n.01	0.93
		(physics) the point of minimum displacement in a periodic system		a system of one or more computers and associated software with common storage	
20	A car that generally has higher performance than classic road cars, in addition to being much more expensive. It is capable of reaching higher speeds than normal, and is often used as a starting point for professional racing competitions.	yacht.n.01	0.12	blind_curve.n.01	0.93
		an expensive vessel propelled by sail or power and used for cruising or racing		a curve or bend in the road that you cannot see around as you are driving	
21	A machine used in agriculture that reaps and threshes cereals in a field	farm_machine.n.01	0.25	combine.n.01	0.85
		a machine used in farming		harvester that heads and threshes and cleans grain while moving across the field	
22	The seasonal movement of livestock, typically moving them between lowlands and highlands during summer and winter	monsoon.n.01	0.18	shortgrass.n.01	0.91
		a seasonal wind in southern Asia; blows from the southwest (bringing rain) in summer and from the northeast in winter		any of various grasses that are short and can tolerate drought conditions; common on the dry upland plains just east of the Rocky Mountains	
23	A structure, usually made of iron, used to mark off-limits areas or maintain order.	crate.n.01	0.23	lock.n.04	0.93
		a rugged box (usually made of wood); used for shipping		enclosure consisting of a section of canal that can be closed to control the water level; used to raise or lower vessels that pass	

	Definizione fornita	Synset ottenuto (lex)	JS	Synset ottenuto (sem)	CS
				through it	
24	Mental content, a representation or concept born from thought, imagination or experience	ghost.n.01	0.33	noun.n.01	0.91
		a mental representation of some haunting experience		a content word that can be used to refer to a person, place, thing, quality, or action	
25	A container generally made of plastic that contains drinkable liquids. It closes or opens with a screwable cap. It also has a label.	opening.n.03	0.14	aerosol.n.02	0.93
		becoming open or being made open		a dispenser that holds a substance under pressure and that can release it as a fine spray (usually by means of a propellant gas)	

Di seguito riportiamo il grafico dei due indici in relazione per visualizzare la qualità delle soluzioni:



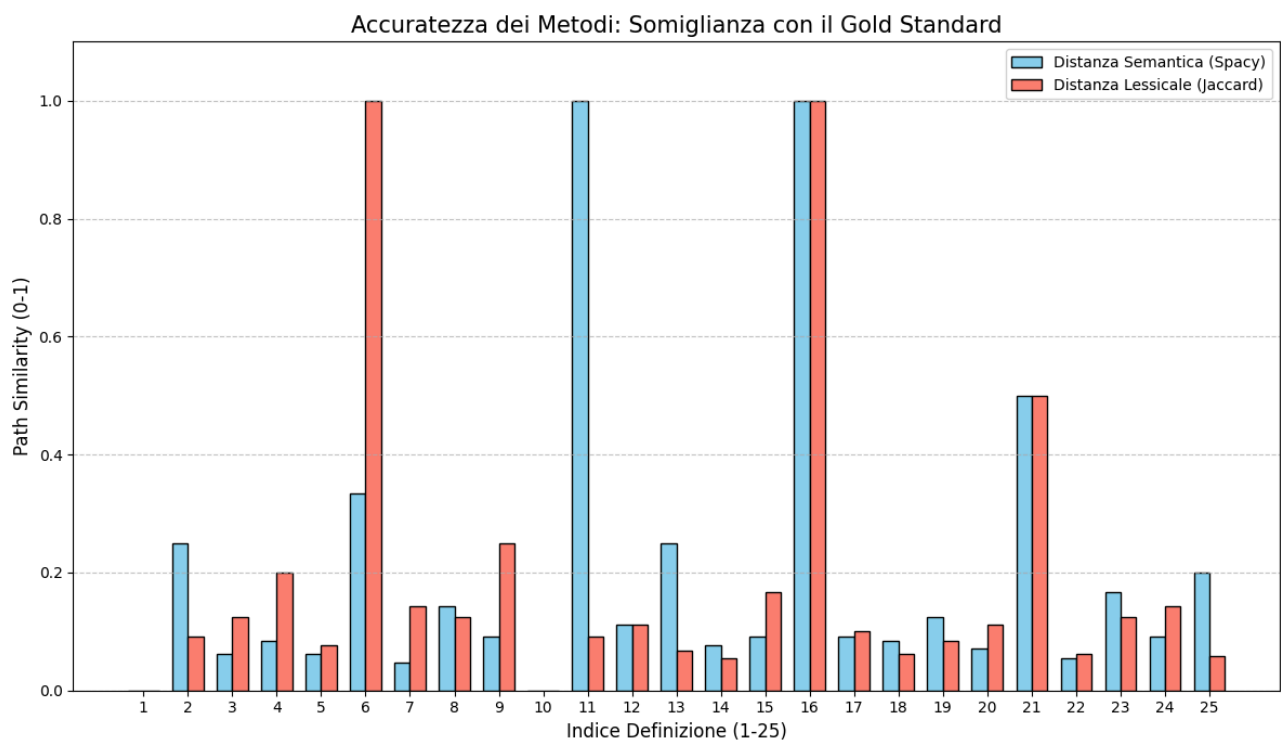
Per avere un metro di paragone, abbiamo elencato i risultati corretti manualmente e li abbiamo confrontati con quelli ottenuti per capire di quanto si discostino da quelli ottenuti. Utilizziamo come misura di accuratezza la Path Similarity: si basa sulla distanza più breve tra i nodi nella gerarchia. Abbiamo escluso dal confronto le definizioni 1 e 10 perchè inesistenti nel WordNet.

Target	Risultato Lessicale	PS	Risultato Semantico	PS
thought.n.03	formation.n.07	0.09	cognition.n.01	0.25
computer_game.n.01	contraception.n.01	0.12	work.n.05	0.06
shame.n.01	brotherhood.n.03	0.20	referred_pain.n.01	0.08
waterworks.n.02	rediffusion.n.01	0.08	electrical_engineering.n.01	0.06
sushi.n.01	sushi.n.01	1.00	rijsttaffel.n.01	0.33
spectacles.n.01	condenser.n.04	0.14	holography.n.01	0.05
gasoline.n.01	resorption.n.01	0.12	helium.n.01	0.14
scientist.n.01	job_candidate.n.01	0.25	reality.n.03	0.09
anger.n.01	emotional_person.n.01	0.09	anger.n.01	1.00
ad.n.01	healthfulness.n.01	0.11	standard_of_living.n.01	0.11
sofa.n.01	kleenex.n.01	0.07	table.n.02	0.25
gambling.n.01	hog.n.02	0.06	age_of_consent.n.01	0.08
screwdriver.n.01	handlebar.n.01	0.17	machine_bolt.n.01	0.09
plunger.n.03	plunger.n.03	1.00	plunger.n.03	1.00
bobby_pin.n.01	sack.n.01	0.10	snood.n.01	0.09
epinephrine.n.01	dressage.n.01	0.06	presence_of_mind.n.01	0.08
push_button.n.01	node.n.04	0.08	computer_system.n.01	0.12
sports_car.n.01	yacht.n.01	0.11	blind_curve.n.01	0.07
harvester.n.02	farm_machine.n.01	0.50	combine.n.01	0.50
migration.n.04	monsoon.n.01	0.06	shortgrass.n.01	0.06
barrier.n.01	crate.n.01	0.12	lock.n.04	0.17

Target	Risultato Lessicale	PS	Risultato Semantico	PS
idea.n.01	ghost.n.01	0.14	noun.n.01	0.09
bottle.n.01	opening.n.03	0.06	aerosol.n.02	0.20

La media per il primo e il secondo metodo di confronto sono rispettivamente:

- Media Semantico: 0.20
- Media Lessicale: 0.19



Riflessioni

Basandoci sui risultati e sulle analisi condotte, possiamo concludere che entrambi i metodi sono poco performanti e inaffidabili data l'accuratezza media ottenuta.

Tuttavia dal punto di vista computazionale, il metodo di confronto semantico consente un'esecuzione più veloce grazie alla possibilità di calcolare una volta sola tutti i vettori dei synset in WordNet. Questo ci consente di eseguire un ciclo più oneroso con la produzione dell'embedding e poi iterare la matrice dei vettori per verificare il più simile a ciascuna entry. Diversamente, utilizzando la misura di Jaccard, siamo obbligati a iterare tutti i synset e calcolare il valore per ogni definizione, rendendo il task evidentemente più lento.

Possiamo ipotizzare che a contribuire all'esito negativo del laboratorio sia stata anche la qualità delle definizioni fornite. Alcune entry hanno descrizioni troppo generiche e altre risultano ambigue.

